

JOBIFY CVUT

AI-Powered Career Opportunity Distribution System

System Architecture Documentation

Developer Guide - Architecture Section

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1. System Architecture Overview

This document describes the system architecture of Jobify CVUT, an AI-powered career opportunity distribution system designed for students at FIT ČVUT. The system follows a layered architecture pattern with clear separation of concerns.

1.1 Architecture Summary

Layer	Components
User Layer	Student User, HR Representative User
Presentation Layer	Web Frontend (HTML/CSS/JS), Discord Bot (JDA)
Application Layer	Spring Boot Backend: Controllers (REST API), Services (Logic), Repositories (JPA), Security (JWT)
Data Layer	PostgreSQL Database, OpenAI API (GPT-4), EXPERTS.AI API

2. High-Level Architecture Diagram

The following diagram illustrates the high-level system architecture showing the layered structure and communication between components.

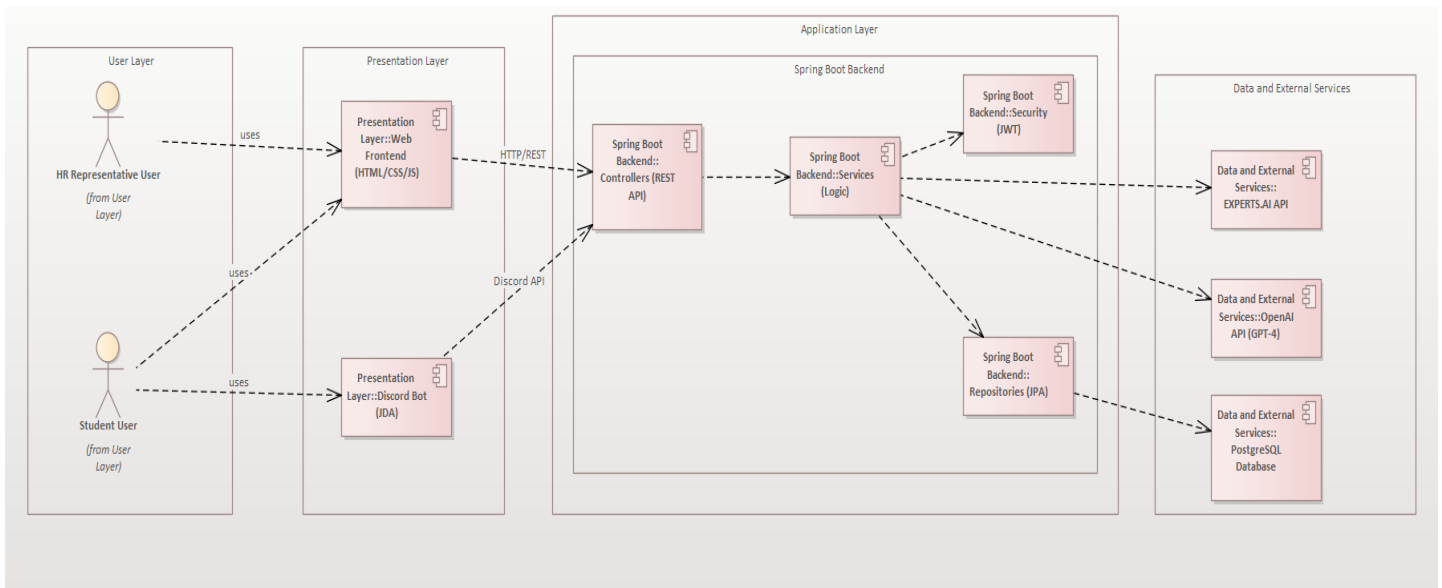


Figure 2.1: High-Level System Architecture

2.1 User Layer

The User Layer represents the two primary actor types that interact with the system:

- **Student User:** FIT ČVUT students seeking internships and job opportunities
- **HR Representative User:** Company recruiters posting job opportunities and reviewing candidates

2.2 Presentation Layer

The Presentation Layer provides two distinct interfaces for user interaction:

- **Web Frontend (HTML/CSS/JS):** Modern responsive web application with dark/light themes, separate dashboards for students and HR
- **Discord Bot (JDA 5.0):** Interactive bot for Discord-based engagement, CV processing, and AI career advice

Communication protocols: HTTP/REST for web frontend, Discord API for bot interactions.

2.3 Application Layer

The Application Layer contains the Spring Boot backend with the following components:

- **Controllers (REST API):** 60+ endpoints handling HTTP requests, routing, and response formatting
- **Services (Logic):** Business logic implementation including job matching, AI integration, and application processing
- **Repositories (JPA):** Data access layer using Spring Data JPA for database operations
- **Security (JWT):** Authentication and authorization using JSON Web Tokens with role-based access control

2.4 Data and External Services

The system integrates with the following data sources and external services:

- **PostgreSQL Database:** NeonDB cloud-hosted relational database with 9 normalized tables
- **OpenAI API (GPT-4):** AI integration for CV review, career advice, and job matching recommendations
- **EXPERTS.AI API:** External job opportunity sourcing from FIT/EDUMATCH feed

3. Use Case Diagram

The Use Case Diagram shows all system functionalities organized by user type and subsystem. It illustrates the interactions between actors (Student, HR Representative) and the system's main features.

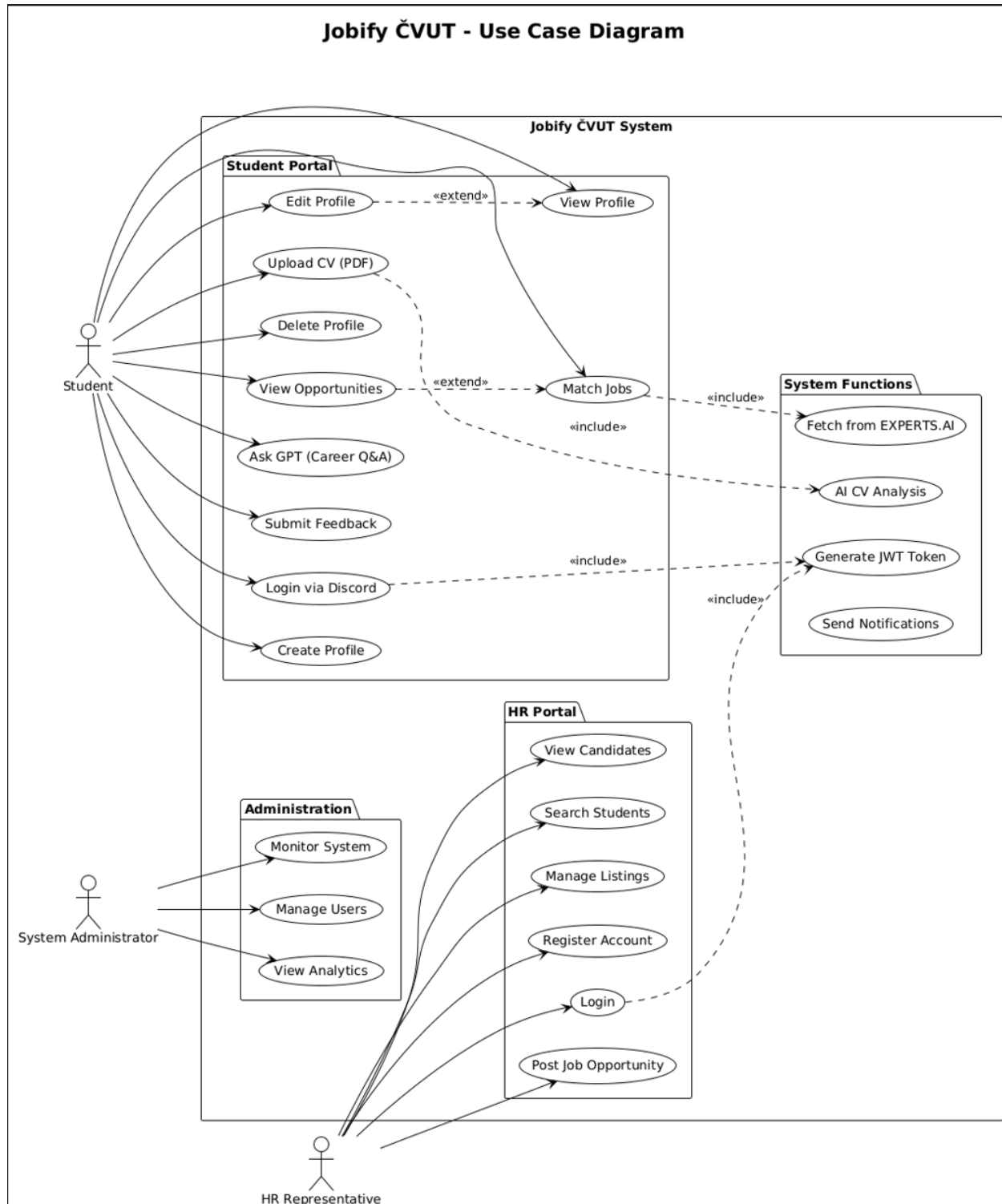


Figure 3.1: System Use Case Diagram

3.1 Use Case Categories

Category	Student Use Cases	HR Use Cases
Authentication	Register, Login, Reset Password, Logout	Register, Login, Reset Password, Logout
Profile	Create/Edit Profile, Upload CV, View Profile	View Student Profiles, Download CVs
Jobs	Browse Matches, Search, Apply, Save, Track	Post Job, Manage Jobs, View Applications
AI Features	Chat with Career Advisor, Get CV Review	Search Talent Pool, View Analytics

4. Activity Diagrams (User Flows)

Activity diagrams illustrate the workflow and user journey through the system for both Student and HR Representative users.

4.1 Student User Flow

The Student User Flow diagram shows the complete journey of a student from registration to job application and career advice.

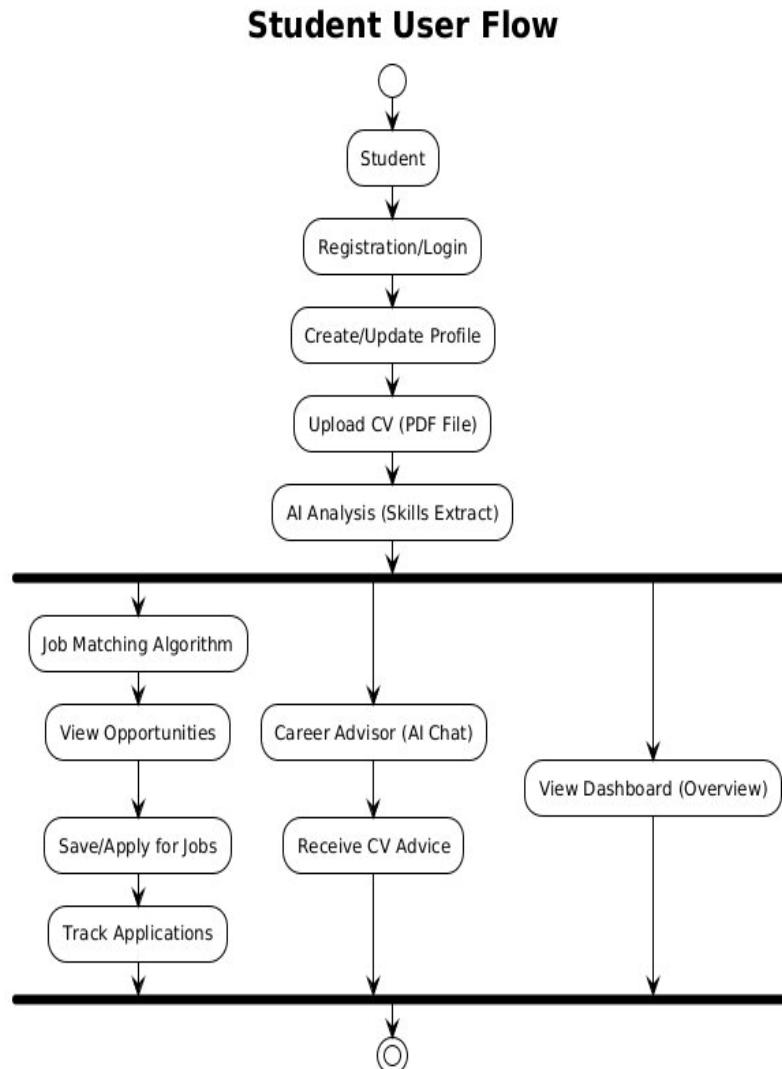


Figure 4.1: Student User Flow Activity Diagram

4.1.1 Student Flow Steps

- **Registration/Login:** Student creates account or logs in with existing credentials
- **Create/Update Profile:** Student fills in personal information, skills, education, and career interests
- **Upload CV (PDF):** Student uploads CV document which triggers AI analysis

- **AI Analysis:** System extracts skills, education, and experience using PDFBox and GPT-4
- **Parallel Activities:** After analysis, student can: browse job matches, use career advisor, or view dashboard
- **Job Matching:** AI algorithm matches student profile with available opportunities
- **Apply/Save Jobs:** Student can save interesting jobs or submit applications
- **Track Applications:** Monitor application status (Pending, Interview, Accepted, Rejected)

4.2 HR Representative User Flow

The HR User Flow diagram shows the workflow for HR representatives managing job postings and reviewing candidates.

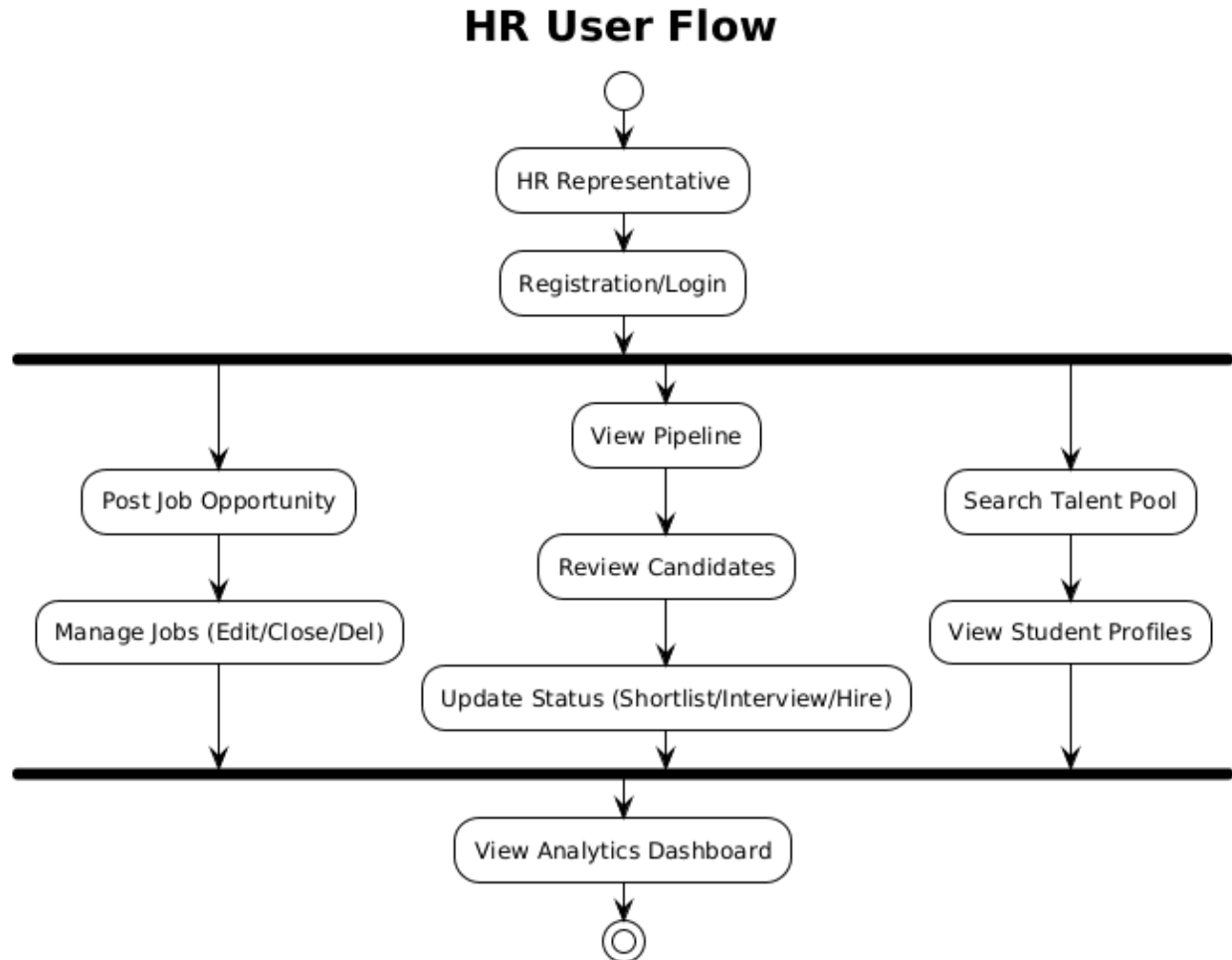


Figure 4.2: HR Representative User Flow Activity Diagram

4.2.1 HR Flow Steps

- **Registration/Login:** HR creates company account or logs in
- **Post Job Opportunity:** Create new job posting with title, description, requirements, and deadline
- **Manage Jobs:** Edit, close, or delete existing job postings
- **View Pipeline:** Review candidate applications in pipeline view
- **Review Candidates:** View student profiles, download CVs, assess qualifications
- **Update Status:** Change application status (Shortlist, Interview, Hire, Reject)
- **Search Talent Pool:** Proactively search for candidates by skills and qualifications
- **View Analytics:** Access HR dashboard with recruitment metrics and statistics

5. Backend Component Architecture

The backend follows a layered architecture with clear separation between controllers, services, and repositories.

Layer	Component	Responsibility
Controllers	AuthController	Student registration, login, token refresh
	HrAuthController	HR registration, login, company management
	OpportunityController	Job listing, search, filtering, details
	AiController	GPT chat, CV review, career advice
Services	MatchService	AI-powered job matching algorithm
	ApplicationService	Job application processing and tracking
Repositories	StudentRepository	Student entity CRUD operations
Security	JwtAuthFilter	JWT token validation and authentication

Jobify - Component Communication Flow

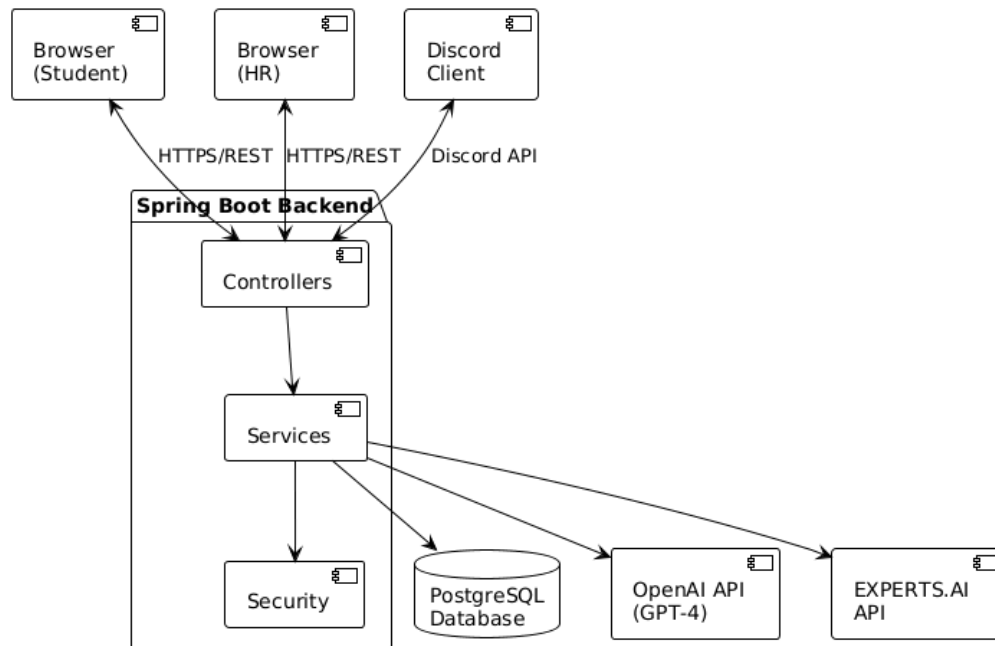


Figure 5.1: Component Communication Flow

6. Technology Stack

Category	Technology	Purpose
Backend Framework	Spring Boot 3.2.0	RESTful API and business logic
Language	Java 17 LTS	Primary programming language
Security	Spring Security + JWT	Authentication and authorization
Database	NeonDB PostgreSQL	Cloud-hosted relational database
Discord API	JDA 5.0.0-beta.23	Discord bot framework
AI Integration	OpenAI GPT-4	CV review, career advice, matching
Frontend	HTML5/CSS3/JS ES6+	Web interface with responsive design
Build Tool	Gradle 8.10	Dependency and build management
CI/CD	GitHub Actions	Continuous integration pipeline